

MTH:3340 ABSTRACT ALGEBRA EXAM 1 STUDY GUIDE

• Recommended Preparation

- Re-work examples from class.
- Work/re-work suggested problems from the book.
- Re-work quiz problems. Understand and learn from mistakes.
- Go over this study guide and make sure you know and understand every point on this study guide.
- Practice showing all your steps. Make solutions organized and be explicit/intentional in your writing.
- Replicate the exam environment. Can you do relevant problems with little to no help?

1.1 Proofs

- Be able to properly form a direct proof.
 - * All variables defined
 - * Formal conclusion
- Be able to properly form a proof by contraposition.
 - * Contrapositive of claim identified
 - * All variables defined
 - * Formal conclusion
- **Be prepared to use these skills in the context of groups.**

Ex. Section 1.4, Practice Problems: 1 - 6
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

 Section 1.4, Additional Exercises: 2 - 4, 8, 9, 13
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

1.2 Sets and Equivalence Relations

- Be able to find the union, intersection of any two sets.
- Be able to find the Cartesian product of sets.
- Be able to define a set using set-builder or roster notation.
- Be able to show that a function is injective, surjective, or bijective.

- Know when a relation is an equivalence relation.
- Be able to show that a relation is an equivalence relation.
- Be able to describe the elements in a particular equivalence class.
- **Be prepared to use these skills in the context of groups.**

Ex. Section 5.1, Exercises: 1 - 13
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

 Section 5.2, Exercises: 7, 8, 12, 21
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

 Chapter 1, Exercises: 1 - 3, 18, 21

2.1 Mathematical Induction

- Be able to prove a statement using mathematical induction.
 - * $P(n)$ defined
 - * Base case shown
 - * Inductive step followed
 - * Formal conclusion
- **Be prepared to use these skills in the context of groups.**

Ex. Section 4.5, Practice Problems: 1 - 6
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

 Section 4.5, Additional Exercises: 2 - 4, 10
 (*Discrete Mathematics: An Open Introduction, 4th Edition*)

 Chapter 2, Exercises: 1 - 5, 9, 10

2.2 The Division Algorithm

- Be able to identify the unique quotient and divisor given two integers a and d with $d > 0$ according to the Division Algorithm.
- Be able to compute the greatest common divisor of two integers.
- Be able to find the Bezout coefficients for a pair of integers a and b . That is, find integers r and s such that $\gcd(a, b) = ar + bs$.
- Be able to determine whether an integer is prime using Trial Division.
- Be able to find the prime factorization of an integer.
- **Be prepared to use these skills in the context of groups.**

Ex. Chapter 2, Exercises: 15 (additionally find their prime factorizations)

3.1 Integer Equivalence Classes and Symmetries

- Know when two integers are congruent modulo some positive integer m .
- Be able to do algebra using the elements in the set $\mathbb{Z}_m$
- Be able to identify the symmetries of a geometric figure.
- Be able to define the rigid motions of a geometric figure using two-row function notation.
- **Be prepared to use these skills in the context of groups.**

Ex. Chapter 3, Exercises: 1

Identify all of the symmetries of a regular pentagon. Define the corresponding rigid motions using two-row function notation.

Identify all of the symmetries of a 45-45-90 triangle. Define the corresponding rigid motions using two-row function notation.

3.2 Definition of a Group

- Know the definition of a group.
- Be able to determine whether a function is a binary operation on a set G .
- Be able to prove that a pair $(G, \circ)$ is a group.
- Be able to prove that a pair $(G, \circ)$ is an abelian group.
- Know how to read and create Cayley tables.

Ex. Chapter 3, Exercises: 2 - 7, 10, 13, 14, 25 - 27, 29, 31 - 33